

Lab 4 - Helicopter | Chat history

Preparation transcript - 1 October 2026

This document records the user-visible messages and clarification answers available during preparation. Tool internals and hidden reasoning are excluded. The implementation was carried out from the supplied repository and lab handout. No separate student prompts for individual tasks were supplied or invented.

User

Attached Lab_4_VibeCoding_Student_handout.pdf and supplied https://github.com/SETAPESU26/08_helicopter

Assistant

I'll read the lab handout and inspect the GitHub repository to identify the required changes and deliverables.

Assistant

I'm using the PDF skill to read the handout and check the submission requirements.

Assistant clarification

Where should I push the finished Lab-4 work? Options: Existing lab repo (Recommended); Prepare files only.

User clarification answer

Existing lab repo (Recommended)

Assistant

The repository requires four tasks: fix movement and screen boundaries, add collision and game over, add distance scoring, and add a one-hit shield. The handout also requires separate commits, two 10-second gameplay videos, and a chat-history document in Lab-4. I'll implement and test the game, then prepare those deliverables.

Assistant clarification

Which personal repository should receive the Lab-4 folder? Options: Lab3-Component-Modelling-Architecture; hostel-maintenance-ticketing-lab1; Prepare files only.

User clarification answer

new

Assistant

I found that the helicopter's position is measured from its centre, so the boundary fix must keep its full body inside the window. I'll also treat a shielded wall crossing as one hit, so the same overlap doesn't immediately cause game over on the next frame. You selected a new personal repository. I'll prepare the finished submission first; the GitHub plugin has no repository-creation tool.

Assistant

All four tasks are implemented in separate commits, and the before video is recorded. I'm checking movement reversal, safe passage through gaps, score freezing and reset, and whether the shield absorbs one obstacle while a later hit still ends the game.

User

Reattached the same handout and supplied https://github.com/SETAPESU26/08_helicopter again.

Assistant

The code passes all 11 tests. I'm now reviewing both 10-second videos and packaging the Lab-4 folder with the code and commit history. The GitHub upload remains pending because you requested a new repository.

Assistant

Both videos are exactly 10 seconds and show actual game frames with scripted controls. The completed game passes all 11 tests, and the package preserves the separate task commits. I'm finishing the chat-history PDF and download package.

Assistant

Completed all four game tasks. All 11 tests passed. Download Lab-4_Helicopter.zip. Includes updated code, two 10-second scripted gameplay videos, chat-history PDF, and separate task commits. GitHub upload is pending. May I use the browser to create Lab4-VibeCoding-Helicopter and upload the work? The GitHub plugin cannot create repositories; browser fallback instructions require your approval.

User

yes

Assistant

I'll create the new personal repository and upload the completed Lab-4 work, preserving the separate task commits.

Assistant

GitHub requires a browser sign-in before I can create the repository. I'll open the secure sign-in prompt; credentials stay out of this chat.

User

continue

Assistant

I'll resume from the GitHub sign-in step and check whether the browser is ready to create the repository.

User

continue

Assistant

The GitHub plugin is connected, but the browser still shows GitHub's sign-in page. I'll check whether the new repository exists; if it does, I can upload through the plugin.

Assistant

The new repository hasn't been created yet. Please use the browser handoff to: 1. Sign in to GitHub. 2. Create an empty repository named Lab4-VibeCoding-Helicopter. 3. Reply done. I can then upload the completed Lab-4 work through the GitHub plugin. The plugin supports uploads but cannot create repositories.

User

done

Assistant

I'll verify the new repository and upload the completed code, videos, and chat-history document with separate task commits.

Assistant

The game is uploaded, with a separate commit for each of the four tasks. I'm adding the videos and updated chat-history PDF, then checking that the files on GitHub match the tested version.

Assistant

Uploaded successfully: Lab4-VibeCoding-Helicopter. The Lab-4 folder contains the completed game, before/after videos, and chat-history PDF. Each task has a separate commit. All 11 tests passed on the uploaded code. For the README's chat-link requirement, add this conversation's ChatGPT Share link.

User

reverify

Assistant

I'll recheck the current GitHub version against every task and submission requirement, including the videos, commit history, restart behaviour, and one-hit shield.

Assistant

The extra checks found a shield bug: after absorbing a hit, it could protect against a second hit on the same wall even after the helicopter had moved into the gap. I'll fix that so only one continuous contact is protected, and add a regression test.

Assistant

The shield fix is uploaded, and all 12 tests pass. I also checked that the game starts, renders, and closes cleanly. I'm refreshing the after video and documents so they match the corrected code. The ChatGPT Share link is still missing from the submission.

Implementation and verification notes

Task 1

Cap speed at 6 pixels/frame. Clear opposing momentum on reversal. Apply braking when keys are released. Clamp the helicopter centre between half its height and screen height minus half its height.

Task 2

Compare the player rectangle against top and bottom wall rectangles. Freeze on collision. Show game over and allow R to restart.

Task 3

Add scrolling speed / 10 metres per active update. Display current and final distance. Restart clears distance.

Task 4

Space enables a blue outlined shield. The first collision consumes it. Ignore only that ongoing wall contact. Leaving the wall, including entering its gap, ends the immunity. Any subsequent hit (including the same wall) requires a new shield activation or ends the game.

Testing

12 unittest cases passed, including all safe integer positions in the test gap, 1,000 updates toward each boundary, immediate reversals, both wall collisions, score freezing, reset, repeated shield contact, reactivation, simultaneous obstacles, rendering, and same-wall re-entry with and without a new shield. The actual main() startup, input loop, rendering and clean exit also passed in a headless environment.

Video method

Both MP4s contain 600 actual Pygame-rendered frames at 60 FPS (10 seconds). Inputs are scripted and the captions disclose that fact. The before video was recorded from an untouched copy before source changes; the after video uses the finished source.

Submission status

The user created [kalalvinayak23/Lab4-VibeCoding-Helicopter](#). The starter, each of the four task changes, the regression checks, and the shield verification fix were uploaded as separate commits through the GitHub plugin. This document and both videos are included in the deliverables. No PR was created against the university repository. The source README requests a conversation share link; it remains to be copied from ChatGPT and added to the repository.

Separate task commits

b92d02e Initialize Lab 4 helicopter submission
94294b8 Initial Helicopter starter
130afa6 Task 1: fix direction reversal, cap speed and clamp both boundaries
4c86fab Task 2: detect wall collisions, freeze game and enable restart
abb39dc Task 3: track scrolling distance, display final score and reset on restart
392ca63 Task 4: add visible reusable one-hit shield with continuous-contact handling
696dcad Add regression checks and running instructions
3cf81f2 Add before/after videos, chat history and Lab-4 submission documentation
424c8d2 Fix shield immunity ending when wall contact stops; test same-wall re-entry